

PROPOSED SOLUTION

eRTMAC-NWIS: the missing institutional-memory layer for drilling operations



NWIS assists the drilling engineer; it does not replace the engineer's decision.

TECHNICAL APPROACH

From scanned reports to explainable risk alerts



DOCUMENT INTELLIGENCE

OCR • PyMuPDF • NLP • NER •
Embeddings



DATA & GEO

Python • PostgreSQL • PostGIS •
Pandas



AI / ML

Semantic Search • Similarity Engine •
XGBoost / LightGBM



APPLICATION

FastAPI • React • Leaflet

END-TO-END PIPELINE



Historical Reports

WCR • DDR • Mud Logs •
PDFs



Document Intelligence

OCR → NLP → Event
Extraction
Depth | Formation | Event |
Severity | Mitigation



Knowledge Repository

Structured historical drilling
memory



Offset-Well Matching

Spatial + Geological +
Operational similarity



Current-Well Analysis

Depth + Formation + Drilling
Parameters



Explainable Risk Alert

Risk + Confidence + Historical
Evidence + Source



Active



Nearest
Formation B X



Most relevant
Formation A ✓

Nearest well ≠ Most relevant well

NWIS combines spatial + geological + operational similarity

FEASIBILITY AND VIABILITY

Buildable with OIL's existing data ecosystem

TECHNICAL FEASIBILITY

- ✓ Python-based modular AI pipeline
- ✓ PostgreSQL + PostGIS for spatial well intelligence
- ✓ OCR / NLP for historical report processing
- ✓ Vector search for semantic retrieval
- ✓ ML models for structured risk prediction
- ✓ FastAPI + React dashboard
- ✓ Compatible with phased eRTMAC integration

DATA FEASIBILITY

Potential OIL sources identified in the problem statement

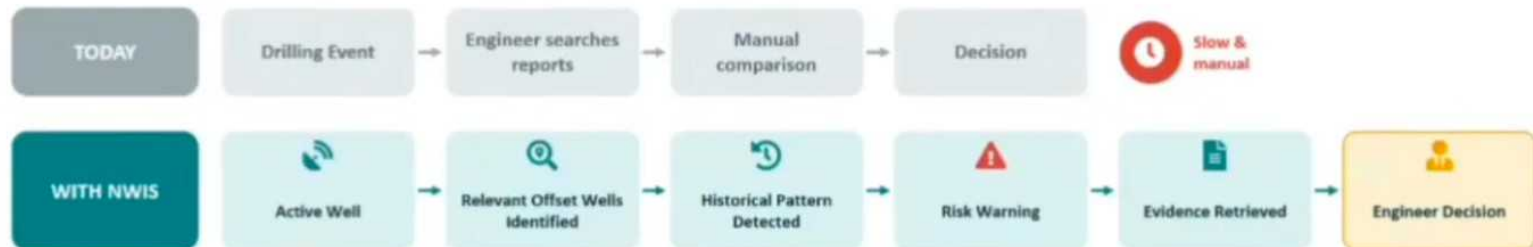


CHALLENGES → OUR STRATEGY



PROTOTYPE FLOW





BENEFITS



INSTITUTIONAL MEMORY

Keeps drilling knowledge beyond individual engineers.



FASTER RETRIEVAL

Find relevant experience without digging through many reports.



CONTEXT-AWARE

Location + formation + depth + trajectory + drilling behaviour.



PROACTIVE RISK AWARENESS

Surfaces historical risk patterns before the well reaches a similar interval.



EXPLAINABLE ALERTS

Why?

Which well?

What happened?

At what depth?

Mitigation?

Which source?

STAKEHOLDERS



Drilling Engineers



Geologists



Operations Teams



Management

From "What happened before?" to "What should we watch for next?"

RESEARCH AND REFERENCES

Existing tools provide pieces of the workflow



Existing Approaches vs NWIS

Parameter	Existing Systems	NWIS
Focus	Detect drilling events after they occur	Forecast risk before it occurs
Time Horizon	Current well only	Current well + historical offset wells
Method	Manual report search	AI: OCR + NLP + ML matching
Basis	Proximity (nearest well)	Formation + trajectory + depth
Output	Log entry / report	Risk alert + historical evidence
Examples	eRTMAC, WCR/DDR archives	eRTMAC-NWIS



KEY REFERENCES

1

SPE DRILLING CONFERENCE

Offset Well Analysis for Well Planning

Argues offset-well data should guide new-well planning to reduce risk from unknown pressures and geology.

onepetro.org/.../482139

2

J. PETROLEUM SCIENCE & ENGINEERING (2021)

Automatic Event Extraction from DDRs

NLP + deep learning extract structured drilling events (kick, stuck pipe, mud loss) from unstructured DDR text.

doi.org/10.1016/j.petrol.2021.108939

3

PETROLEUM & PETROCHEMICAL ENG. J. (2022)

Offset Wells Data Analysis for HPHT Wells

Manual offset-well risk correlation for one HPHT field improved drilling performance and planning.

doi.org/10.23880/ppej-16000298

4

UPSTREAM OIL & GAS TECHNOLOGY (2021)

Data-Driven Indicators for Stuck-Pipe Events

Real-time ML indicators built from mud log data predict stuck-pipe risk on the active well.

doi.org/10.1016/j.upstre.2021.100043

You Tube Link :